C5 Kinetics – Knowledge Organiser

AQA A-level Chemistry • Set 1: Collision Theory and Maxwell–Boltzmann **Paper 2**

#	Question	Answer
1	Rate of reaction	Change in concentration per unit time
2	Activation energy (E_a)	Minimum energy for a reaction to occur
3	Two requirements for a successful collision	Energy ≥ E_a and correct orientation
4	Catalyst	Increases rate without being used up
5	How a catalyst increases rate	Alternative route with lower E_a
6	x-axis of a Maxwell–Boltzmann distribution	Energy
7	y-axis of a Maxwell–Boltzmann distribution	Number of molecules
8	Area under the Maxwell–Boltzmann curve	Total number of molecules
9	Does the curve touch the x-axis at high energy?	No
10	Does the curve start at the origin?	Yes

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AQA A-level Chemistry • Set 2: Factors and Measuring Rate **Paper 2**

#	Question	Answer
1	Five factors affecting rate	Temperature, concentration, pressure, surface area, catalyst
2	Peak of a Maxwell–Boltzmann curve	Most probable energy
3	Position of mean energy relative to the peak	To the right
4	Peak at higher temperature	Lower and further right
5	Area under the curve at higher temperature	Unchanged
6	Effect of a catalyst on the curve shape	None
7	Rate from a concentration–time graph	Gradient of tangent
8	Mean rate equation	Change in quantity ÷ time
9	Units of rate	mol dm^{-3} s^{-1}
10	Why a 10 °C rise has a large effect on rate	Many more particles with energy ≥ E_a